

Part I: Theory Questions

(Chapter-wise Collection of All Theory Questions)

Module 1: Diodes and Circuit Applications

1. **(W-24, S-22, S-20, S-23)** Explain the working of P-N junction diode in forward and reverse bias conditions with energy band diagram. Draw its V-I characteristics.
2. **(S-25, W-19, S-23)** Define P-N junction diode. Explain V-I characteristics of P-N junction diode.
3. **(W-24, W-22, S-19)** Explain Full Wave Rectifier (Center-tapped and Bridge) with neat circuit diagram and waveforms. Derive expressions for efficiency and ripple factor.
4. **(W-24, S-25, W-21)** Explain Half Wave Rectifier with neat diagram and draw input-output waveforms. Derive expression for efficiency and ripple factor.
5. **(S-25, W-23)** Compare Half wave, Full wave Center-tapped and Bridge rectifiers.
6. **(W-24, S-25, S-22)** Explain Series and Shunt Clipper circuits with suitable diagrams and waveforms.
7. **(S-24, W-19)** Explain Clamper circuits (Positive and Negative) with circuit diagrams and waveforms.
8. **(W-24)** Draw and explain Voltage Multiplier (Doubler/Tripler) circuit.
9. **(S-25, S-20)** Derive the expression for Load Line Analysis of a diode circuit.
10. **(W-22)** What is a Filter? Explain Capacitor filter with Full wave rectifier.
11. **(W-21, S-24)** Explain Diode Approximations (Ideal, Practical, Complete).
12. **(S-24)** Explain the effect of temperature on the V-I characteristics of a PN junction diode.
13. **(S-24)** Draw circuit diagram of positive clamper and explain the operation.
14. **(S-24)** Explain biased positive and negative clipper circuits in detail.

Module 2: Special Purpose Diodes

1. **(W-24, S-25, W-21, S-20)** Explain Zener Diode as a Voltage Regulator with circuit diagram. Differentiate between Zener and Avalanche breakdown.
2. **(W-24, S-25, W-22)** Explain construction and working of Solar Cell. Draw its I-V characteristics.
3. **(S-25, W-19, S-24)** Explain construction and working of LED (Light Emitting Diode). List its applications.

4. (W-24, S-25) Write a short note on Tunnel Diode. Explain its V-I characteristics.
5. (S-25) Explain construction and working of Varactor Diode.
6. (S-25) What is Schottky Diode? Explain its advantages over conventional diodes.
7. (S-24, W-23) Explain construction and operation of Photodiode.
8. (S-24) Explain working of PIN photodiode.
9. (W-22) Explain Laser Diode working principle.

Module 3: Bipolar Junction Transistor (BJT)

1. (W-24, S-25, W-19, S-24) Explain Input and Output characteristics of CE, CB, and CC configurations. Compare them.
2. (W-24, S-25) Derive the relationship between α and β current amplification factors.
3. (W-24, S-22) Explain various Biasing methods: Fixed Bias, Collector-to-Base Bias, and Voltage Divider Bias. Derive stability factor for each.
4. (S-25, W-21, S-24) Explain the concept of DC Load Line and Q-point. Discuss the factors affecting stability of Q-point (Thermal Runaway).
5. (W-24, S-24) Draw and explain the hybrid (h-parameter) model for a transistor in CE configuration.
6. (W-24) Explain BJT as a switch and BJT as an amplifier.
7. (W-22) Explain Early Effect (Base width modulation).
8. (S-24) Explain thermal stability of BJT with stability factors S, S', S'' .
9. (S-24) Draw Emitter follower circuit and discuss its input and output characteristics.
10. (W-18) Explain Transistor current components.
11. (S-24) Draw Hybrid equivalent model of transistor for CB, CE, CC configurations.

Module 4: Field Effect Transistors (FET)

1. (W-24, S-25, S-19) Explain construction and working of N-channel and P-channel JFET with characteristics.
2. (W-24, S-25, W-22) Explain construction and working of D-MOSFET and E-MOSFET (Depletion & Enhancement types) with diagrams.
3. (W-24, S-25, S-24) Differentiate between BJT and FET / JFET and MOSFET.
4. (S-25) Explain Voltage Divider and Self Biasing of JFET.
5. (W-24) Explain Transconductance (g_m) and Amplification factor (μ) in FET.
6. (W-24, S-24) Explain FET as an amplifier.

7. **(S-24)** Describe MOSFET operation and draw the structures for D-MOSFET & E-MOSFET.
8. **(S-20)** Explain Pinch-off voltage in JFET.
9. **(W-21)** Define FET parameters: Drain resistance, Transconductance, Amplification factor. Derive relation between them ($\mu = r_d \times g_m$).

Module 5: Digital Electronics

1. **(W-24, S-25, S-24)** Write a short note on Seven Segment Display.
2. **(S-24, S-23)** Explain Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) with truth tables and symbols.
3. **(S-24)** Realize AND, OR, NOT gates using Diodes.
4. **(S-22)** Explain Universal Gates (NAND, NOR) and implement basic gates using them.
5. **(W-19)** Convert the following numbers: Binary to Decimal, Decimal to Binary, Hex to Binary.
6. **(W-22)** Explain De Morgan's Theorems.
7. **(S-20)** Explain Full Adder and Half Adder circuits.

Part II: Numerical Problems

(Chapter-wise Collection of All Numerical Problems)

Module 1: Diodes (Numericals)

1. **(W-19)** A crystal diode having internal resistance $r_f = 20\Omega$ is used for half wave rectification. If the applied voltage is $v = 50 \sin \omega t$ and load resistance $R_L = 800\Omega$, find: (i) I_m , I_{dc} , I_{rms} (ii) DC Output Voltage (iii) Efficiency of rectification.
2. **(S-22)** In a full wave rectifier, the input is from a 30-0-30 V transformer. The load resistor is $R_L = 2k\Omega$. Each diode has a forward resistance of 10Ω . Calculate: (a) dc voltage (b) dc current (c) ripple factor (d) efficiency.
3. **(W-21)** For a Zener diode regulator, determine the range of input voltage for which output voltage remains constant. Given: $V_z = 10V$, $R_s = 1k\Omega$, $R_L = 2k\Omega$, $I_{zmin} = 5mA$, $I_{zmax} = 50mA$.

Module 3: BJT (Numericals)

1. **(S-24)** For a BJT, α_{dc} is 0.98 and base current is $50 \mu A$, then find β_{dc} , I_c & I_E .
2. **(W-22)** For a CE transistor amplifier, $V_{CC} = 12V$, $R_C = 2k\Omega$. The Q-point is $(6V, 3mA)$. Determine the value of Base resistor R_B if $\beta = 100$. (Fixed Bias Circuit).
3. **(S-23)** A transistor has $\beta = 150$. Calculate the approximate collector and base currents if $I_E = 12mA$. Also find α .
4. **(W-19)** For a voltage divider bias circuit, $V_{CC} = 20V$, $R_1 = 10k\Omega$, $R_2 = 2.2k\Omega$, $R_C = 3.6k\Omega$, $R_E = 1k\Omega$. Assume silicon transistor. Calculate V_{CE} and I_C .

Module 5: Digital Electronics (Numericals)

1. **(W-22)** Convert: (i) $(11011)_2 = (?)_{10}$ (ii) $(45)_{10} = (?)_2$ (iii) $(A2B)_{16} = (?)_2$
2. **(S-20)** Perform binary addition and subtraction: (i) $10110 + 11001$ (ii) $11011 - 01101$



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